David Nguyen

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Education

University of California, Irvine - B.S. in Computer Science - *Dean's Honor List 2020-2024* **2020 - 2024**
University of Texas at Austin - M.S. in Computer Science **2026 - 2028**

Experience

Software Engineer Intern - Tricia **April 2025 - July 2025**

- Developed and maintained responsive web application features using React and TypeScript, improving UI consistency and performance across multiple components
- Integrated Supabase as a backend-as-a-service solution, handling authentication (Google OAuth), real-time database interactions, and API routing
- Participated in debugging and performance tuning of frontend and backend components, resulting in faster load times and improved user experience

Technical Projects

Infotainment **March 2025 - April 2025**

- Developed a feature-rich system with a modern UI designed in Figma, integrating multimedia playback and connectivity features using ReactJS.
- Built a robust backend with Node.js and Express, handling API requests such as authentication (OAuth) and token refreshes efficiently
- Created 18 components featuring geopositioning using Zippopotamus API, temperature forecasting using Open-Meteo API, immersive video using the YouTube iFrame player API, and Spotify integration (listening to music, searching, and browsing playlists) using the Spotify Web API.

DAMP Search Engine **April 2024 - May 2024**

- Launched a scalable search engine with Python, HTML, CSS, and JavaScript capable of handling 55,000+ documents
- Built an inverted index using BeautifulSoup, indexing over 2,000,000 tokens to efficiently answer search queries under 100ms using cosine similarity and PageRank
- Enhanced content summaries using the Phi3 LLM for improved query relevance

Fabflix **September 2023 - December 2023**

- Deployed a full-stack RESTful API web application following a microservice architecture that allows users to browse, search, and checkout 1000+ movies using Java, JDBC, MySQL, Tomcat, HTML, CSS, JavaScript, and AWS
- Developed ~20 servlets with features such as cart checkout backed by sessions, secure login using SHA256 hashing, full-text search, auto-complete backed by a cache, and bot detection using reCAPTCHA
- Improved website performance by 150% through optimizations including MySQL connection pooling, MySQL replication, and Apache load balancing

Mission Control: All Systems Go - NASA & ASU (Capstone Project) **January 2023 - June 2023**

- Led the development of 1 of 4 minigames: "Among the Stars" using C# and Unity as part of a NASA & ASU collaboration
- Applied agile methodologies and feedback loops to deliver high-quality, stakeholder-approved results
- Reduced the release cycle from weekly to daily by streamlining the CI/CD process to enhance deployment

Skills

Programming Languages: Python, C/C++, Java, C#, JavaScript, TypeScript, SQL
Frameworks & Tools: Figma, React, FastAPI, AWS (EC2), Docker,
Databases: MySQL, PostgreSQL, Cassandra, MongoDB, Couchbase, Neo4j
Other: Git, Agile Development, RESTful API Design
Soft Skills: Analytical thinking, teamwork, communication, adaptability, attention to detail, problem-solving